

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 34)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

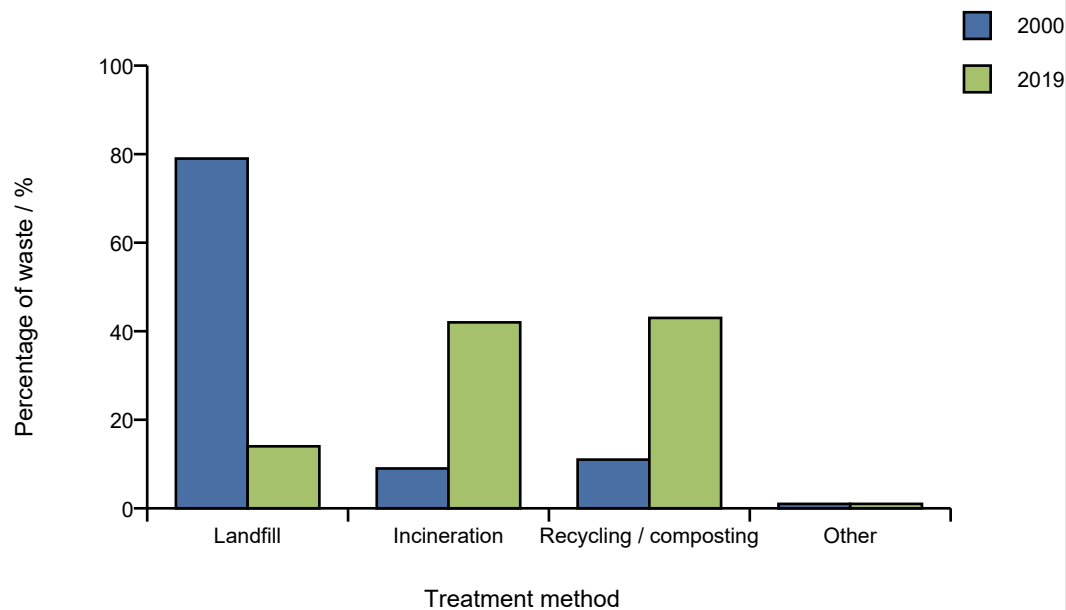
Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1

. 1

Describe the changes shown in Figure 1.

[2 marks]

0 1

. 2

Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

1

2

3

0 1

. 3

Describe **four** features of a well-managed landfill site.

[4 marks]

1

2

3

4

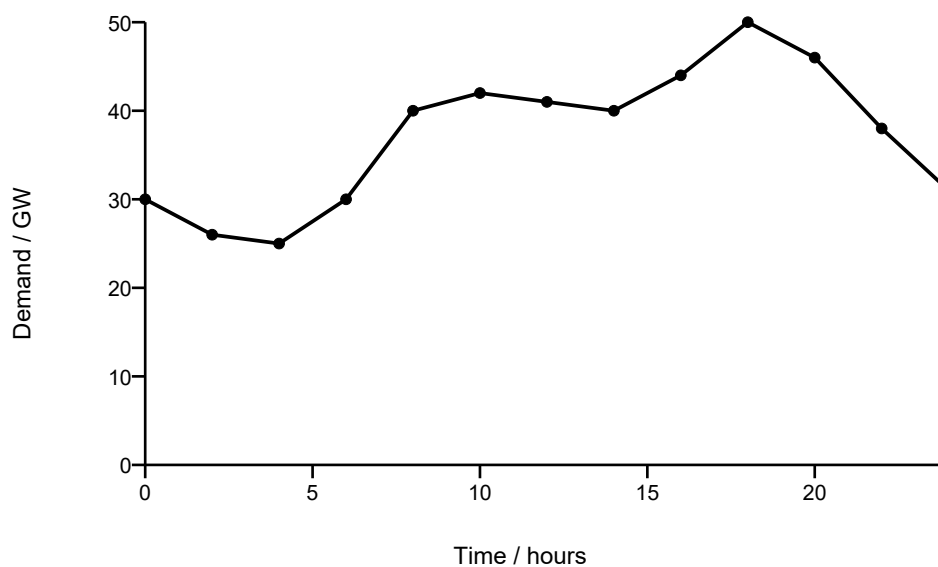
0 1 . 4 Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

10

0 2 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 2 . 1 Describe the pattern of demand shown in Figure 1.

[2 marks]

0 2 . 2

Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

0 2 . 3

Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

0 2 . 4

Give **two** causes of fluctuations in energy **supply**.

[2 marks]

1

2

10

03 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location

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Emission timing

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03 . 2

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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03 . 3

Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 4

Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0 4

1

Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

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0 4

2

Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

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0 4 . 3 State **two** changes that would decrease the cut-off ore grade. [2 marks]

1

2

10

0 5 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 5 . 1 Calculate the estimated population size. [2 marks]

Answer beetles

0 5 . 2 State **four** assumptions of the Lincoln Index method. [4 marks]

1

2

3

4

0 5 . **3** Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

0 5 . **4** Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

10

Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

Complete Table 1 by explaining how each feature reduces oil pollution.

Do not write outside the box

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes

GPS / Automatic Identification System

06 . 3

Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

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06 . 4

Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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10

07

Table 1 shows the water quality required for four different uses.

07 . 1

Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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07 . 2

Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

07 . 3

Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

07 . 4

Give **two** methods used to sterilise treated water.

[2 marks]

1

2

10

08 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

- A High sand content
B High clay content
C Crumb peds
D High pH

☐
☐
☐
☐

08 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

08 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

09 . 1

Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

09 . 2

Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

09 . 3

Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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2

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3

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09 . 4

Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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10

10 . 1

Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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Extra space

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10 . **2**

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

END OF QUESTIONS